DIOPHANTINE OBSTRUCTIONS, AFFINE ORTHOGONAL RUNGE THEOREMS, AND DEGREE-512 EXTRASPECIAL COVERS FOR MULTIPLICITY FIVE IN PASCAL'S TRIANGLE

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with Erdős Problem #849 on integers occurring with multiplicity at least five in Pascal's triangle, we establish four unconditional arithmetic and Galois-theoretic results. First, any nontrivial collision $\binom{n}{k} = \binom{m}{\ell}$ with $5 \le \ell < k \le n/2$ satisfies $K(n-k) \ge 8$ and $\omega(n-k) \ge 7$ (and $\omega(n-k) \ge 14$ for $\ell \in \{5,6,7,8\}$). Second, for every $g \ge 3$, the affine orthogonal group $G_g = \mathbb{F}_2^{2g} \rtimes O^+_{2g}(2)$ first admits an intransitive subgroup at index $2^{2g} - 2^g$. Third, on the genus-four $(5,6)$ collision curve C, the two-power θ-compatible multiplication tower achieves the full affine Kummer holomorph image. Fourth, descending to the nonabelian pro-2 fundamental group quotient, we construct the pointed degree-512, genus-1537 extraspecial cover $(Z, z_x) \to (C, x)$ with geometric deck group $E = 2^{1+8}_+$ and prove via Griess's nonsplit extension $1 \to \mathrm{Inn}(E) \to \mathrm{Aut}(E) \to O^+_8(2) \to 1$ that its arithmetic pole permutation image is the full holomorph $\mathrm{Hol}(E) = E \rtimes \mathrm{Aut}(E)$, obstructing the direct Runge pole condition over every number field K.

1. Main Mathematical Results

Theorem 1.1 (Pascal multiplicity-five prime-divisor bounds and extraspecial Runge obstruction)**.** (1) *Every nontrivial solution to* $\binom{n}{k} = \binom{m}{\ell}$ *with* $5 \le \ell < k \le n/2$ *satisfies* $K(n-k) \ge 8$ *and* $\omega(n-k) \ge 7$*; for* $\ell \in \{5,6,7,8\}$*,* $\omega(n-k) \ge 14$*.*

(2) *Uniformly for* $g \ge 3$*, the affine orthogonal group* $G_g = \mathbb{F}_2^{2g} \rtimes O^+_{2g}(2)$ *has minimum intransitive subgroup index* $2^{2g} - 2^g$*.*

(3) *On the genus-four* $(5,6)$ *curve* $C\colon (T-1)(T-9)(T-25) = 384u(u^2-1)(u^2-4)$*, the pointed degree-*512*, genus-*1537 *extraspecial cover* $(Z, z_x) \to (C, x)$ *with geometric deck group* $E = 2^{1+8}_+$ *has full arithmetic pole permutation group* $\mathrm{Hol}(E) = E \rtimes \mathrm{Aut}(E)$*, forcing* $r_Z(K) \le r_1(K) + r_2(K)$ *for every number field* K*.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2338 in the domain of Diophantine equations and algebraic number theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Unconditional omega(a) >= 7, K(a) >= 8, and omega(a) >= 14 on 5,6,7,8, sharp affine orthogonal Runge theorem for G_g = F_2^(2g) semidirect O^+_(2g)(2), two-power theta-tower holomorph image, and degree-512 genus-1537 nonabelian extraspecial 2_+^(1+8) Runge obstruction.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Complete classification of all integers appearing at least five times in Pascal's triangle

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References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 2338 (Diophantine equations and algebraic number theory), archived in the accompanying repository package `results/2338/`.